

UTS-UAV: A Context-Grounded Benchmark for Operational Event Discovery in Low-Altitude UAV Imagery

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Abstract—Operational UAV inspection requires more than classifying a supplied crop: a system must reject ordinary frames, find a region that contains sufficient evidence, and recognize the event under severe scale and deployment shifts. We introduce UTS-UAV, a real low-altitude UAV benchmark with three defining properties: (1) *operational realism*, pairing event-bearing frames with true no-event negatives from the same acquisition process; (2) *context-grounded supervision*, linking each positive full frame to a human-accepted contextual ROI, its high-resolution crop, and an event label; and (3) *deployment-oriented evaluation*, with leakage-controlled session-disjoint, unseen-site, and forward-temporal protocols. Five controlled tasks isolate event presence, contextual grounding, oracle-region recognition, joint discovery, and cascaded deployment. A controlled 4,778-image session-disjoint study shows that a shared Qwen3-VL classifier reaches 72.24% macro-F1 on ground-truth crops and 71.98% with ROI plus context, but falls to 70.11%, 68.64%, and 67.34% with a drawn ROI, masked background, and coordinates, respectively. We further propose Feature-Space Qwen-Ground (Qwen-Ground), which grounds an event directly from one full-image Qwen visual encoding. A multi-scale feature decoder predicts presence and a contextual box; feature-space ROIAlign then combines local and global evidence with definition prototypes for classification. The benchmark registers closed-set, open-vocabulary, generative, cascade, calibration, robustness, transfer, and code-backed ablation comparisons. On the frozen session-disjoint test set, Qwen-Ground obtains 98.96% presence AP, 3.68% contextual AP50, 39.72% positive-class macro-F1, and 3.92% grounded mAP50. Direct generative Qwen reaches 6.10% grounded mAP50 but yields 46.61% no-event false positives, versus 16.61% for Qwen-Ground. Adaptive tiling reduces false positives to 3.75% while retaining 57.68% positive recall. These contrasting operating points expose calibration and contextual localization as unresolved deployment bottlenecks.

I. INTRODUCTION

Low-altitude UAVs can survey farmland, waterways, construction zones, and settlements at a scale that is impractical for ground patrols. A deployable inspection system must answer three coupled questions: *is* an event present, *where* is the evidence-bearing region, and *what* event does it represent? The first question is easily hidden by positive-only datasets; the second is difficult because useful evidence may occupy a tiny fraction of a 4K frame; and the third is contextual because labels such as exposed waste, unclean water, or active construction describe conditions, not pixel-tight object categories.

Existing aerial datasets are rich but leave this combination under-specified. DOTA and VisDrone emphasize generic objects under scale and viewpoint variation [1], [2]; ERA and CapERA study event recognition or description from selected clips [3], [4]. Recent UAV vision–language benchmarks greatly broaden semantic, spatial, and planning evaluation [5]–

[7], while AirZoo contributes geometry-rich supervision and cross-domain transfer tracks [8]. These advances do not directly quantify the failure chain from ordinary-scene rejection to context-region discovery and fine-grained operational recognition. A crop classifier can therefore appear deployment-ready without ever searching a full frame or facing a true negative.

The closest recent grounding benchmarks sharpen this gap. AerialVG and DVGBench test explicit or implicit references in high-resolution drone imagery [9], [10]; UniRef-UAV additionally admits absent targets and variable output cardinality [11]. Their targets remain referred objects. In operational inspection, by contrast, the query is implicit (“find a reportable event”), the positive region may be a diffuse condition, and the first correct output for many frames is no-event.

We build UTS-UAV from a production inspection archive while treating its workflow provenance as part of the benchmark contract. The source platform connects mission dispatch and multi-source image ingestion to AI-assisted retrieval, operator review, alerting, replay, and reporting. We distill that stream rather than its proprietary interface: full-frame/crop pairs are linked, contextual boxes are recovered and manually accepted, operational labels are frozen, and genuine no-event frames are retained. Content, session, and site groups are assigned before splitting, and test labels are sealed. The benchmark consequently supports controlled counterfactuals: a crop estimates oracle recognition performance; a ground-truth ROI tests whether full-frame compression and background are harmful; presence and class-agnostic grounding measure search; and joint or cascaded inference measures the perception front end of an alert–review–report loop.

Following the property–track organization of large dataset papers and the capability–diagnostic structure of recent UAV benchmarks, we make four contributions:

- **A context-grounded operational benchmark.** UTS-UAV combines true positive and no-event frames with full-resolution imagery, contextual ROIs, crops, event labels, and leakage-safe acquisition groups.
- **Five interoperable tasks and three shifts.** A single record contract supports presence-aware grounding, two oracle recognition settings, unified discovery, and cascaded discovery under session, site, and temporal generalization.
- **A controlled representation study.** On the same 4,778 session-disjoint positive records, we compare crop, contextual, drawn-box, coordinate, and masked views with one shared Qwen adaptation, alongside ConvNeXt,

Operational event discovery: reject, ground, and recognize

Positive frames contain a contextual evidence ROI; genuine no-event frames require an empty prediction.

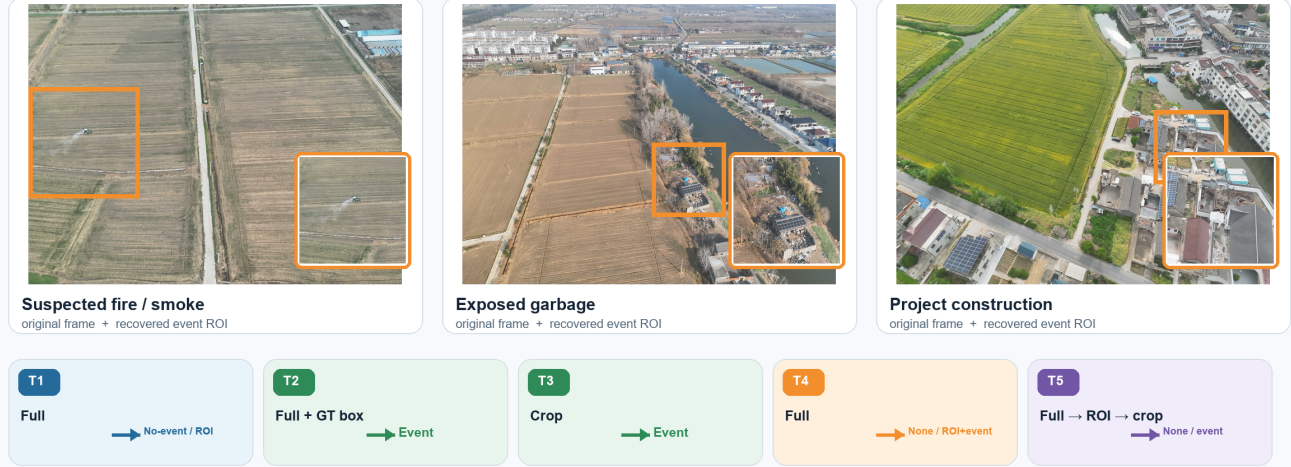


Fig. 1. UTS-UAV evaluates full-frame operational event discovery. Positive examples (shown) link a full frame, a human-accepted contextual ROI, and a crop; no-event frames have an empty ROI target. The five tasks isolate presence/localization, oracle-region recognition, joint prediction, and a modular cascade. Boxes denote sufficient contextual evidence rather than pixel-tight physical objects.

DINOv2, and Florence-2 references.

- **A full-frame feature-space baseline and diagnostic suite.** Qwen-Ground couples Qwen visual tokens to multi-scale contextual grounding, explicit presence prediction, feature-space ROI pooling, and definition-prototype classification. Fourteen registered analyses cover calibration, scale, long tails, evidence, robustness, efficiency, and transfer.

II. RELATED WORK

Aerial perception and operational events. DOTA and VisDrone established aerial detection under extreme scale, orientation, and density variation [1], [2]; UAV-Human and MAVREC extend aerial understanding to human behavior and unscripted multi-view recognition [12], [13]. ERA and CapERA target aerial event recognition and captioning [3], [4], whereas A2Seek studies reasoning-centric video anomaly localization [14]. UTS-UAV focuses on static, high-resolution operational inspection and explicitly couples true no-event rejection, contextual region grounding, and event recognition.

UAV foundation-model benchmarks. GeoChat introduced grounded region interaction for remote sensing [15]. UAVBench/UAUIT-1M evaluates ten image- and region-level tasks and provides large-scale instruction tuning [5]; MM-UAVBench organizes 19 subtasks around perception, cognition, and planning [6]; and SpatialUAV uses 14 task types, seven input configurations, and nine answer formats to diagnose low-altitude spatial intelligence [7]. SpatialSky-Bench further exposes navigation-specific geometric weaknesses [16]. AirZoo organizes aerial retrieval, cross-view matching, and reconstruction around large-scale geometric pretraining and real-domain transfer [8]. Inspired by their systematic taxonomies, we organize one operational decision into five linked tasks and ask where the discovery chain fails.

Newest UAV suites further emphasize grounded, inspectable answers. UAVReason unifies aerial reasoning and dense generation [17]; UAV-DualCog requires spatial or temporal grounding and reports a human reference [18]; and UAVQA-Bench evaluates tool-routed training-free agents with intermediate verification and difficulty-adaptive search [19]. RSVideo couples small-target evidence selection with background-token compression in remote-sensing videos [20]. Our static protocol isolates representation, adaptation, and search under sealed no-event/ROI/event outputs.

Grounding, detection, and unified vision models. Grounding DINO grounds arbitrary text in boxes [21], YOLO-World provides efficient open-vocabulary detection [22], and D-FINE improves real-time DETR localization [23]. Florence-2 expresses detection and grounding in a prompt-driven sequence representation [24]; RegionGPT and Qwen3-VL strengthen region and coordinate reasoning [25], [26]; DINOv2 supplies transferable visual features [27]. Qwen3-VL-Seg adds a lightweight mask decoder conditioned on MLLM box priors [28]; our visual-only proposal path instead learns no-event/context-ROI outputs directly. AerialVG exposes the high-resolution and relation-reasoning bottleneck [9], DVGBench targets implicit references [10], and UniRef-UAV explicitly measures no-target discrimination [11]. GrabVG separates hypothesis search from graph-based relational binding [29]. Operational events differ from canonical objects: their regions can be diffuse and their names need not describe a single visible entity. We therefore compare closed-set, open-vocabulary, and generative formulations, and introduce a context-aware no-event model rather than assuming that every frame contains a target.

Leakage, calibration, and deployment shift. Adjacent

UAV frames and repeated site visits are highly correlated. Random image splits can reward memorization of place, weather, or capture trajectory. Our content-, session-, and site-grouped positive protocols, forward-time test, sealed labels, and registered group-wise uncertainty analysis make generalization part of the benchmark definition. Negative-frame calibration is evaluated jointly with localization because a useful inspection system must trade recall against false alarms rather than optimize positive-only accuracy. Aerial Mirage documents hallucination in aerial LVLM outputs [30]; accordingly, invalid strings, unsupported event claims, and plausible boxes on negatives remain scored failures in our evaluator.

III. UTS-UAV BENCHMARK

A. Operational System Provenance

The source archive was exported, under company authorization, from a production inspection platform rather than assembled by web search. Scheduled and ad-hoc missions feed extracted video frames, image libraries, and local uploads into an AI-assisted alert–review–report loop. This creates correlated captures, ordinary frames that never become alerts, and positives whose meaning depends on task and scene context. The operational taxonomy even separates similar material by setting—for example land, water, collection-point, and bank-side garbage, or construction- versus factory-side storage. Such distinctions motivate contextual ROIs: a tight debris or material box can erase the evidence that defines the reportable condition.

B. Three Benchmark Properties

Operational realism. UTS-UAV contains **TBD** full-resolution frames, comprising **TBD** event-bearing samples, **TBD** true no-event samples, and **TBD** operational event classes. Negatives are captured by the same operational process rather than synthesized from random background patches, enabling meaningful false-alarm and calibration analysis.

Context-grounded supervision. A sample is

$$d_i = (I_i, p_i, b_i, C_i, y_i, m_i), \quad (1)$$

where I_i is the full frame, $p_i \in \{0, 1\}$ is event presence, b_i is a contextual evidence ROI, C_i is its high-resolution crop, y_i is the event label, and m_i contains opaque acquisition groups. For $p_i = 0$, $(b_i, C_i, y_i) = \emptyset$; for $p_i = 1$, the full/ROI/crop views share one record identity. The ROI encloses evidence sufficient for an operator to judge the event. It can include surrounding water, terrain, buildings, or activity context and is not a pixel-tight physical-object boundary.

Deployment-oriented protocols. Content, acquisition session, site, and time groups are frozen before model development. This supports controlled evaluation on repeated operations, unseen geography, and future capture periods without allowing near-duplicate frames to cross a boundary.

C. Construction and Quality Assurance

Positive full/crop pairs are linked using delivery keys and media hashes. Crop-to-frame registration proposes each ROI using local features, geometric verification, and corner projection; every proposed box is subsequently accepted or corrected by human review. Negatives are verified as containing no benchmark event under the frozen ontology. Label adjudication retains the source operational distinctions, with a supported-class macro average and a separate rare-class report. Company authorization, privacy review, release redaction, and test-label sealing are completed. Public records expose only opaque session/site identifiers and the metadata necessary to reproduce the protocols.

D. Five Controlled Tasks

T1: presence-aware contextual grounding maps a full frame to either no-event or one ROI. **T2: oracle-box recognition** supplies a positive full frame and its GT ROI as drawn, coordinate, or masked-background prompt. **T3: oracle-crop recognition** classifies the delivered positive crop and supplies an oracle reference. **T4: unified discovery** returns no-event or an ROI–event pair in one call. **T5: modular discovery** first proposes no-event/ROI and then classifies a source-resolution crop. T2–T3 isolate representation and context; T1 isolates presence/search; T4–T5 compare unified and modular deployment while sharing the same records.

E. Leakage-Controlled Protocols

The primary **session-disjoint** protocol assigns entire acquisition and content groups to train, validation, or test (**TBD/TBD/TBD** frames). **Unseen-site** holds opaque geographic groups out for test (**TBD/TBD/TBD**), and **forward-temporal** trains on past captures and tests on later months (**TBD/TBD/TBD**). Positive records obey the corresponding content/session/site/time constraint. The 2,534 no-event frames use one fixed group-disjoint partition (1,630/344/560, with 3/1/3 source groups) shared by all three protocols. Thus negative evaluation is group-disjoint, but we do not attribute site or temporal semantics to negative retention until aligned metadata are available. Every derived task keeps the parent record assignment, and all model, threshold, prompt, and context-margin selection is validation-only.

F. Metrics and Statistical Protocol

Presence reports AP, AUROC, F1, and negative-frame false-positive rate (N-FPR) at a validation-selected operating point; calibration reports ECE and Brier score. On positive frames, contextual localization reports AP and recall at IoU $\{0.25, 0.50, 0.75\}$, mean IoU, and normalized center error. Lower IoU thresholds are retained because the annotation is a sufficient-context region, not a physical-object extent. Recognition reports supported-class macro-F1, accuracy, and tail macro-F1.

End-to-end contextual grounding AP ($G\text{-mAP@}\tau$) treats a prediction as a match only when its event label is correct and $\text{IoU}(\hat{b}, b) \geq \tau$; predictions on no-event frames are false positives. We also report positive-frame joint success

TABLE I

POSITIONING AGAINST REPRESENTATIVE AERIAL BENCHMARKS. “NO-EVENT” MEANS EXPLICIT ORDINARY/NEGATIVE FRAMES ARE PART OF THE TARGET EVALUATION; CONTEXT ROI DENOTES A SUFFICIENT-EVIDENCE REGION RATHER THAN A PIXEL-TIGHT OBJECT BOX. SHIFT REQUIRES GROUPED GEOGRAPHIC OR TEMPORAL TESTING.

Benchmark	Main scope	Real UAV	Region output	No-event	Context ROI	Full→event discovery	Site/time shift
ERA [3]	event recognition	✓	–	–	–	–	–
VisDrone [2]	object detection/tracking	✓	✓	–	–	–	–
A2Seek [14]	video anomaly reasoning	✓	✓	✓	–	✓	–
UAVBench [5]	image/region VL	✓	✓	–	–	–	–
AerialVG [9]	relational grounding	✓	✓	–	–	–	–
UniRef-UAV [11]	universal referring	✓	✓	✓	–	–	partial
MM-UAVBench [6]	perception–planning	✓	partial	–	–	–	–
SpatialUAV [7]	spatial intelligence	✓	✓	–	–	–	–
UTS-UAV	operational events	✓	✓	✓	✓	✓	✓

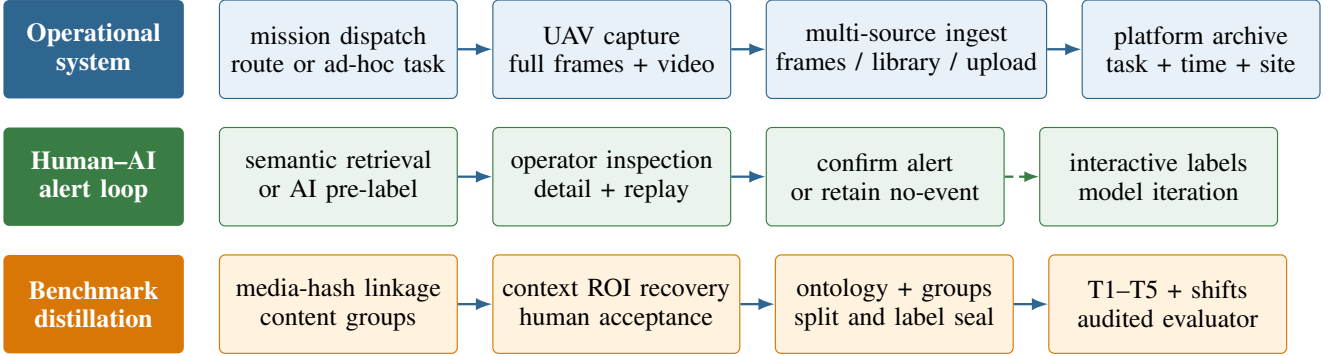


Fig. 2. From production inspection to a public benchmark. The source system organizes mission-driven acquisition, heterogeneous ingestion, AI-assisted retrieval, operator review, alerting, and iterative labeling. UTS-UAV distills this stream into hash-linked records, accepted contextual ROIs, verified no-event examples, frozen groups, and sealed evaluation tasks. The diagram is an original abstraction; proprietary interfaces and identifiers are not redistributed.

TABLE II

FIVE LINKED TASKS. POS.: POSITIVE EVENT FRAME; NEG.: NO-EVENT FRAME.

Task	Input	Output	Pos.	Neg.
T1	full frame	no-event or ROI	✓	✓
T2	full + GT ROI	event class	✓	–
T3	GT crop	event class	✓	–
T4	full frame	no-event or ROI+event	✓	✓
T5	full→pred. crop	no-event or ROI+event	✓	✓

$J@τ$, valid-output rate, parameters, peak memory, throughput, and median/P90 latency. Completed single-seed experiments report point estimates; paired group-bootstrap intervals are reported only for conditions with aligned record-level outputs and never substituted for training-seed variation. Slices are fixed before test evaluation: ROI-area quartile, class frequency, boundary clarity, time period, site, and negative subtype.

Because IoU alone cannot establish that a contextual box carries the decision evidence, we add a frozen *evidence-sufficiency* diagnostic. A committee of T3 recognition probes measures macro-F1 when given only the proposed ROI (F_{in}) and after that ROI is removed from the full frame (F_{out}). The evidence-sufficiency gap $ESG = F_{in} - F_{out}$ is reported with each grounding system, but is not folded into G-mAP. This counterfactual test rewards regions that are both sufficient

and consequential without pretending their edges are object boundaries.

IV. FEATURE-SPACE QWEN-GROUND

Qwen-Ground uses the visual tower from the Qwen3.6-35B-A3B-FP8 checkpoint; direct generative baselines use Qwen3-VL-8B [26]. It separates three questions that generative coordinate strings entangle: whether an event exists, where its sufficient context lies, and which event definition best matches the grounded evidence. The registered input cap is 995,328 pixels. The language decoder is never called.

A. Multi-Scale Contextual Grounding

Let F_q denote spatial tokens after Qwen’s native merger. A learned projection maps them to 768 dimensions. Adaptive average pooling forms a three-level pyramid at scales $s \in \{1, 2, 4\}$; two-dimensional coordinate and level embeddings retain location and scale:

$$F_s = \text{AAP}_s(W_q F_q) + E_{xy,s}. \quad (2)$$

A single event query cross-attends to the concatenated pyramid through a one-layer, 12-head Transformer decoder [31]. On the finest grid, cosine similarity between the decoded event feature z_e and token f_i defines a spatial distribution a_i . The box center is a soft-argmax; width and height use the

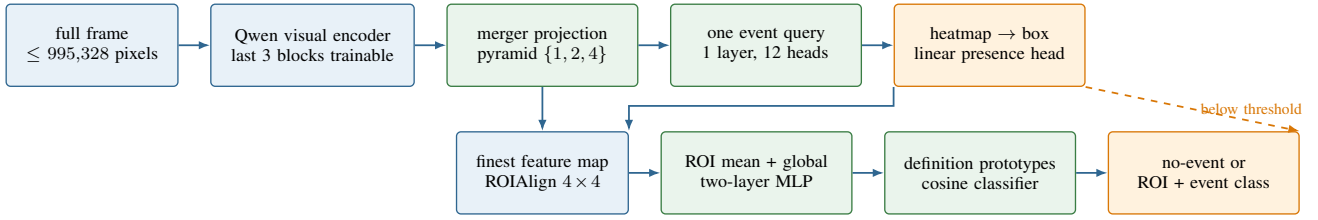


Fig. 3. Qwen-Ground processes the full frame once. Qwen merger tokens feed a multi-scale event-query decoder for presence and contextual-box prediction. ROIAlign reuses the finest feature map; no source-pixel crop, language decoding, adaptive tiling, or counterfactual verifier is part of the model.

event feature and heatmap-weighted visual evidence:

$$\begin{aligned}
 a_i &= \text{softmax}_i(\tau \cos(z_e, f_i)), \\
 \hat{c} &= \sum_i a_i p_i, \\
 (\hat{w}, \hat{h}) &= \sigma \left(g_b[z_e, \sum_i a_i f_i] \right).
 \end{aligned} \tag{3}$$

A separate linear head gives event presence $\hat{p} = \sigma(w_p^\top z_e)$. This is one event query plus a presence head, not a null/event query bank.

B. Feature-Space ROI Recognition

The predicted box indexes the existing finest feature map through 4×4 ROIAlign. Spatial averaging yields a local feature z_r without re-encoding source pixels. A two-layer MLP combines z_r with the global event feature. Its normalized projection is compared by cosine similarity with 18 fixed class-definition prototypes t_k :

$$\begin{aligned}
 z_r &= \text{Avg}(\text{ROIAlign}(F_1, \hat{b}, 4, 4)), \\
 \ell_k &= \gamma \cos(W_{cg}[z_e, z_r], t_k).
 \end{aligned} \tag{4}$$

Removing either z_r or z_e is therefore an interpretable ablation of local or global evidence. A fixed linear classifier replaces the prototype head in a separate control.

C. Optimization and Inference

Localization uses $5\mathcal{L}_1 + 2\mathcal{L}_{\text{GIoU}}$ and a spatial-KL loss on the heatmap; presence uses focal loss and positive records additionally use cross-entropy classification:

$$\mathcal{L} = \mathcal{L}_{\text{focal}} + p(5\mathcal{L}_1 + 2\mathcal{L}_{\text{GIoU}} + \mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{CE}}). \tag{5}$$

Training first establishes multi-scale grounding and then attaches the recognition branch. The final three vision blocks are updated; earlier vision blocks stay frozen. Registered learning rates are 10^{-6} for the trainable vision blocks and 10^{-4} for new heads. Positive sampling is class balanced, and batches expose 20% no-event records. Box-conditioned classification uses GT boxes in epochs 1–3, predicted detached boxes with probabilities 1/3 and 2/3 in epochs 4 and 5, and predicted detached boxes thereafter. The current protocol trains 12 epochs with seed 43 and evaluates the saved final checkpoint; it does not claim early stopping or multi-seed uncertainty.

At inference, the validation-frozen presence threshold returns either no-event or one canonical $(\hat{b}, \hat{y})$ record. T1

exposes presence and box outputs; T4 adds the feature-space classifier. T2/T3 remain oracle-representation tasks, and T5 is an exposed modular cascade rather than another head of the same model.

V. EXPERIMENTS

Benchmark construction, contextual-box acceptance, governance, and test-label sealing are complete. Black numbers below were recomputed from saved per-record predictions. Red cells reserve results that still require the corrected frozen evaluator; they are not estimates.

A. Models, Training, and Reproducibility

The comparison spans **oracle classifiers** (ConvNeXt, DINOv2 [27], Florence-2 [24], and Qwen3-VL [26]); **closed-set localizers** (Faster R-CNN, RT-DETR, DINO, and D-FINE [23]); **open-vocabulary grounders** (Grounding DINO [21] and YOLO-World [22]); generative Florence-2/Qwen3-VL; and modular T5 cascades. AerialVG, DVG-Bench, UniRef-UAV, and UAV open-vocabulary pretraining provide transfer initializations [9]–[11], [32].

Every run uses the same record IDs and ontology. Table III is the completed session-disjoint seed-43 evaluation; it is not presented as a three-seed mean. The shared Qwen representation model is Qwen3-VL-8B-Instruct with LoRA rank 16, $\alpha = 32$, dropout 0.05, five epochs, batch size 4, gradient accumulation 2, and a 995,328-pixel cap. One mixed-view sampler trains the five Qwen views; the epoch selected on validation maximizes their mean macro-F1 (epoch 4). Classification uses the final non-padding LLM hidden state and a learned head, not generated text, so 100% validity is structural. All future multi-seed or bootstrap summaries will be reported only after their registered replicates exist. Direct T4 Qwen uses a separate three-epoch, rank-16 adapter with constrained category/location tokens; adaptive tiling reuses it. Saved timings have different batch sizes and scopes (Table IV); matched batch-one end-to-end timing remains pending.

B. RQ1: How Much Does the Input Representation Hide?

On the same 4,778 positives, Qwen reaches 72.24 macro-F1 with a GT crop and 71.98 with ROI plus 20% context, a difference of only 0.27 points. A drawn ROI reaches 70.11, while masked background and coordinate-only conditioning fall to 68.64 and 67.34. Thus region metadata does not substitute for visually focused evidence, and removing all surroundings can discard useful event context. The result is

TABLE III

SESSION-DISJOINT ORACLE-REPRESENTATION RESULTS ON THE SAME 4,778 POSITIVE RECORDS (SEED 43). THE FIVE QWEN ROWS SHARE ONE QWEN3-VL-8B LORA ($r = 16$) AND HIDDEN-STATE CLASSIFIER; WEIGHTS ARE FIXED ACROSS VIEWS. CTX. IS A 20% MARGIN. ALL ENTRIES ARE PERCENTAGES.

Model	Positive input	Acc.	Macro-F1	Valid
ConvNeXt [33]	GT crop (T3)	87.00	56.88	100
DINOv2 [27]	GT crop (T3)	90.08	66.39	100
Florence-2 [24]	GT crop (T3)	91.00	68.55	100
Qwen3-VL [26]	GT crop (T3)	91.38	72.24	100
Qwen3-VL [26]	full + drawn ROI	90.90	70.11	100
Qwen3-VL [26]	full + coordinates	90.10	67.34	100
Qwen3-VL [26]	masked background	90.27	68.64	100
Qwen3-VL [26]	ROI + ctx.	91.13	71.98	100

a controlled single-seed representation finding, not evidence about localization quality or uncertainty across training runs.

C. RQ2: Can Models Reject Negatives and Ground Positives?

T1 compares global, tiled, and feature-pyramid inputs; closed-set and open-vocabulary pretraining; and validation-only threshold calibration. FROC and precision-recall curves pair positive recall with false alarms on 560 held-out no-event frames. Localization is sliced by ROI area, diffuseness, border contact, class frequency, and negative subtype. Qwen-Ground uses class-balanced positive sampling and 20% no-event exposure; it does not use online hard-negative mining. D-FINE gives the lowest completed N-FPR (2.14%) but recalls only 0.92% of positives at that operating point. Grounding DINO reaches 90.14% P-AP but only 0.11% AP50. YOLO-World fires on every negative at its validation-selected threshold and has AP50 below 0.01%. These results separate presence ranking from a deployable rejection/localization operating point and show that open-vocabulary transfer alone does not solve contextual event grounding.

D. RQ3: Unified Model or Modular Cascade?

T4 and T5 share presence, grounding, recognition, and efficiency metrics. The cascade exposes specialist errors; Qwen-Ground reuses one visual feature map for search and classification. Matched-compute views cap pixels, visual tokens, and wall-clock budget. Errors are decomposed into false alarm, miss, localization-only, classification-only, joint, and invalid-output outcomes. Direct Qwen leads completed rows in P-AP (99.41%), AP50 (5.14%), C-F1 (44.80%), and G-mAP50 (6.10%), but its validation-selected threshold produces 261 false alarms among 560 negatives (46.61%). Qwen-Ground gives 3.92% G-mAP50 and 93 false alarms (16.61%), with 81.62% positive recall. Its median positive IoU is 0.233 and Recall@0.5 is 21.77%, so contextual localization remains weak. All G-mAP50 values now use the same single-candidate presence-score ranking; the former all-class-score diagnostic for Qwen-Ground is excluded from the primary comparison.

Adaptive tiling inspects the global image, searches overlapping 2×2 tiles below a fixed 0.85 confidence, and reinspects the best source-pixel crop. It averages 2.63 visual calls (maximum six). Its own validation threshold gives 21 false alarms (3.75% N-FPR) but only 57.68% positive recall; G-mAP50 falls to 5.52% and C-F1 to 36.54%. Additional views therefore do not uniformly improve discovery. The D-FINE-DINOv2 cascade’s 23.70% candidate C-F1 yields only 0.89% G-mAP50, illustrating the effect of missed localization.

E. RQ4: Does Performance Transfer Across Place and Time?

A session-selected shortlist is frozen before unseen-site and forward-temporal evaluation. Positive results report absolute score and relative retention. The shared no-event split is group-disjoint but lacks protocol-aligned site/time semantics; therefore cross-site/time negative retention is marked pending and will not be interpreted as geographic or temporal transfer. *Projected trend—verify and replace after the frozen run: Report positive-record transfer and, separately, session-level negative false alarms; do not merge them into a semantic cross-shift claim.*

F. RQ5–RQ8: Calibration, Scale, Tail, and Format

Calibration (RQ5) reports reliability, ECE, Brier score, N-FPR@95% recall, and risk-coverage at validation-frozen cutoffs. **Scale (RQ6)** crosses model capacity, 640/1024/1536-pixel inputs, fixed tiling, native dynamic resolution, memory, and latency. **Tail and geometry (RQ7)** crosses head/medium/tail classes and reweighting with ROI size, diffuseness, and boundary contact. **Prompt/format (RQ8)** compares zero/few-shot and class-definition prompts with JSON, coordinates, and multiple choice; invalid output remains a failure. *Projected trend—verify and replace after the frozen run: Insert only registered sweeps and their Pareto fronts; state explicitly when a change affects calibration but not ranking.*

G. RQ9: Which Qwen-Ground Components Matter?

The frozen one-change ablations remove heatmap-guided center/feature estimation, multi-scale pooling, ROI or global evidence, and the GT-to-predicted-box curriculum, or replace definition prototypes with a linear classifier. Matched configurations inherit 12 epochs and a 10^{-4} head learning rate; legacy eight-epoch controls are separate. Both training stages share each geometric ablation. *Projected trend—verify and replace after the frozen run: Report AP50, C-F1, and G-mAP50 only after all seven runs complete; make no directional claim from an incomplete ablation grid.*

H. Registered Extended Diagnostics

The supplement registers data-scale and few-shot curves, ROI sensitivity and image-random leakage controls, evidence-sufficiency probes, physical corruptions [34], aerial transfer, and visual-call budgets. These diagnostics keep validation/test fixed and do not alter the primary benchmark claims; results remain unreported until the required frozen runs and replicate seeds are complete.

TABLE IV

PRIMARY SESSION-DISJOINT RESULTS ON 4,778 POSITIVE AND 560 NO-EVENT TEST RECORDS (SEED 43). P-AP: EVENT-PRESENCE AP; N-FPR: FALSE-POSITIVE RATE ON NO-EVENT FRAMES AT THE VALIDATION-SELECTED THRESHOLD; AP50: CONTEXTUAL LOCALIZATION AP; C-F1: CANDIDATE CLASSIFICATION MACRO-F1 ON POSITIVE RECORDS BEFORE PRESENCE GATING; G-MAP50: SINGLE-CANDIDATE EVENT-AWARE AP RANKED BY PRESENCE SCORE FOR EVERY SYSTEM. SCORES EXCEPT TIME ARE PERCENTAGES. TIME IS THE RECORDED PER-IMAGE COST, WITH HETEROGENEOUS SCALES, NOT A SPEED RANKING. RED CELLS AWAIT THEIR FROZEN RUN.

System	Family/mode	P-AP↑	N-FPR↓	AP50↑	C-F1↑	G-mAP50↑	ms
D-FINE [23]	closed-set T1	84.86	2.14	2.29	–	–	162.3
Grounding DINO [21]	open-vocab. T1	90.14	5.18	0.11	–	–	834.8
YOLO-World [22]	open-vocab. T1	89.20	100.00	<0.01	–	–	139.3
Qwen3-VL [26]	generative T4	99.41	46.61	5.14	44.80	6.10	685.5 ^a
Florence-2 [24]	generative T4	88.33	6.07	0.19	4.40	0.04	340.2
D-FINE + DINOv2 [23], [27]	modular T5	84.86	2.14	2.29	23.70	0.89	246.6
Best loc. + Qwen3-VL [26]	modular T5	TBD	TBD	TBD	TBD	TBD	TBD
Qwen adaptive tiling [26]	global/tiles/crop	98.57	3.75	4.76	36.54	5.52	1725.9 ^c
Qwen-Ground (ours)	feature-space T4	98.96	16.61	3.68	39.72	3.92	79.7 ^b
Human reference	stratified subset	TBD	TBD	TBD	TBD	TBD	–

^aBatch-4 preprocessing/generation cost per image, excluding decoding/I/O. ^bBatch-8 transfer/forward cost per image, excluding loader/postprocessing. ^cSum of cached-global, batched-tile, and crop-call costs; routing/I/O excluded. Matched batch-one end-to-end timing is pending.

TABLE V

FROZEN DEPLOYMENT-SHIFT SHORTLIST. RET.: G-MAP50 RETENTION RELATIVE TO THE SESSION-DISJOINT SCORE; ALL THRESHOLDS REMAIN FIXED.

System	Session G-mAP50	Unseen site G-mAP50 / Ret.	Future time G-mAP50 / Ret.
Best generative VLM	TBD	TBD/TBD	TBD/TBD
Best cascade	TBD	TBD/TBD	TBD/TBD
Qwen-Ground	TBD	TBD/TBD	TBD/TBD
Human reference	TBD	TBD/TBD	TBD/TBD

VI. BENCHMARK DESIGN AND IMPLICATIONS

Why a 0/1 contextual ROI formulation? Each record asks whether an event is visible and, if so, which contextual region supports it. The formulation penalizes unconditional positive output and preserves operator-defined context. Exhaustive multi-instance and multi-event evaluation remain separate extensions.

How should contextual grounding be interpreted? A contextual ROI defines an evidence budget, not a segmentation boundary. AP/R at 0.25/0.50/0.75, normalized center error, ROI-area slices, and jitter robustness jointly test neighborhood search and tolerance to boundary variation. With weights fixed, strong crop but weak oracle-box recognition implicates compression or prompting; strong T2/T3 but weak T4 motivates search/calibration diagnosis, not an automatic causal attribution.

IoU cannot establish causal evidence. ESG tests whether retaining the ROI preserves recognition and removing it degrades recognition, with random-area and saliency-matched controls. Because ESG depends on frozen recognition probes, it remains diagnostic; agreement across two probes and the human sufficiency subset is reported explicitly.

Why does platform provenance matter? The source system is organized around missions, alerts, replay, and reports, not independent images. This explains the need for

grouped splits, no-event records, and context-sensitive labels. Figure 2 exposes that origin without showing the proprietary interface; T4/T5 evaluate its perception front end, not dispatch, remediation, or case closure.

Leaderboard organization. Three separate tracks preserve distinct operating questions: *Oracle Recognition* reports T2/T3 macro-F1 and tail behavior; *Presence and Grounding* reports T1 P-AP, N-FPR, contextual AP, and calibration; *End-to-End Discovery* reports T4/T5 G-mAP, validity, and latency. A single average could hide false alarms behind easy crop accuracy. G-mAP, N-FPR, latency, and memory define a Pareto view across the three deployment protocols. The supplement fixes adaptation/data-access tracks, model/input budgets, output-repair rules, sealed receipts, and anytime curves.

Implications for operational deployment. No-event is a semantic prediction, not a review decision. A separate validation-frozen selector routes uncertain outputs to review while retaining all records in primary scores. Risk-coverage, residual misses, and review load are tested under site/time shift without assuming perfect human correction; the intended use is decision support. Since missions contain correlated frames, image-level N-FPR is necessary but not sufficient; deduplicated false alerts per mission/hour are reported only when duration metadata permit them.

VII. LIMITATIONS, ETHICS, AND INTENDED USE

Each positive record supplies one primary contextual ROI; the benchmark does not measure exhaustive instances or segmentation. Diffuse ROI edges motivate multi-threshold, center-error, and jitter analyses instead of a single strict IoU. The ontology is region-informed, rare classes remain difficult, unseen sites do not imply global coverage, and images do not evaluate tracking, causal explanation, or autonomous action.

The frame benchmark does not reproduce temporal alert clustering, live replay, dispatch, or report closure, so N-FPR is not itself operator workload. Proprietary interfaces, accounts,

routes, and coordinates are not released; Figure 2 is a sanitized workflow abstraction.

The data support perception research and decision support, not autonomous enforcement. Authorization, privacy/figure review, and test-label sealing are complete. Public artifacts use opaque site/session IDs and exclude precise routes, coordinates, and unnecessary identifiers; protected test annotations remain private. Models may encode geographic or socioeconomic bias, so results must be read by site, time, class, and false-alarm slice rather than as a universal safety claim.

VIII. CONCLUSION

UTS-UAV links no-event rejection, contextual grounding, and recognition across five tasks and three deployment shifts. ROI plus context nearly matches GT-crop Qwen macro-F1 (71.98 versus 72.24), whereas coordinates alone lose 4.90 points. On the frozen session-disjoint test, direct Qwen reaches 6.10% G-mAP50 with 46.61% N-FPR; Qwen-Ground reaches 3.92% with 16.61% N-FPR. Adaptive tiling reaches 5.52% G-mAP50 and 3.75% N-FPR but retains only 57.68% positive recall. These results reveal a substantial gap between oracle recognition and reliable event discovery across operating points.

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